



SOFTWARE QUALITY AND RELIABILITY

Semester 7th – 9th

Academic Year 2023 – 2024

STATEMENT OF RELATIVE WORK – PART B

A. Development of a Function Library (DLL) for Personnel Management

As part of the development of the Human Resources Information System of a company, you are asked to develop a .dll library named **HRLib**, which will contain a series of methods and functions for the management of the company's personnel. Specifically, the description of the functions is as follows:

1. **bool ValidName(string Name)**

This function accepts a text as a parameter and returns a true or false value if the parameter can correspond to an employee's first and last name or not, respectively.

2. **bool ValidPassword(string Password)**

This function accepts a text as a parameter and returns true or false if the parameter is an acceptable password. Password requirements: a. At least 12 characters, b. A combination of uppercase letters, lowercase letters, numbers and symbols (at least one character of each), c. The letters must be Latin characters, d. It must start with a capital letter and end with a number.

3. **EncryptPassword(string Password, ref string EncryptedPW)**

*This method accepts a password as a parameter and returns the encrypted code based on Caesar's Cipher, with ASCII alphabet (128 characters, see [here](#)) and sliding by 5 positions. The _____
This result is returned through the corresponding by ref parameter it accepts.*

4. **CheckPhone(string Phone, ref int TypePhone, ref string InfoPhone)**

This method accepts a number as text and checks if it corresponds to a phone. The results are returned via the corresponding by ref parameters it accepts. Specifically, if it is a landline phone, it returns zero (0) as the phone type (TypePhone) and as the phone information (InfoPhone) the zone to which the phone number belongs (8 zones, see [here](#)). If it is a mobile phone, it returns the number one (1) as type and the mobile phone company as information (see [here](#)). Otherwise, if it is not a valid phone number, it returns the number type -1 and null in the information.

5. **InfoEmployee(Employee EmplX, ref int Age, ref int YearsOfExperience)**

This method accepts as parameters the details of an employee, and returns his age and years of service in the company. These results are returned through the corresponding by ref parameters it accepts.



6. int LiveInAthens(Employee Empls)

This function returns the number of employees residing in Athens.

This function calls the CheckPhone function.

In addition, this library will contain the following data structure (struct):

1. **Employee** with fields: string Name (employee name), string HomePhone (landline), string MobilePhone (mobile phone), datetime Birthday (date of birth), datetime HiringDate (date of hiring).

B. Creating Test Cases (TestCases)

For each method and function of the library, the corresponding tables with the control cases should be created:

id	Entry Condition	Equivalent price classes	
		Valid	Invalid

Per.El.	Test Input Data *				Expected Results *	
#id	Variable1	Variable2	Variable3	Result1	Result2	

* They contain as many columns as there are input data and expected results, respectively.

C. Developing Unit Tests (UnitTests)

For each method and function of the library, corresponding unit control methods should be developed.

D. Audit reports

The purpose of these reports is to record in detail the events that occur during the execution of the controls which indicate the existence of errors, in order to contribute to their effective correction.

Therefore, you should record at least 5 error reports with the following structure:

1. Error ID (consisting of a verbal error identifier and a sequential number)
2. Name of test method that failed
3. Description of the test performed (i.e. the test method)
4. Report an error that occurred
5. Screenshots of Test Explorer



DELIVERABLES

The submission will be made via eclass, in the Assignments section, where a team member is asked to attach the following in zip format:

1. Workbook (text document in .pdf format) which will be structured per implementation method/function and for each one will contain its implementation assumptions (*Required A*), the tables of control cases (*Required B*), the implementation of the corresponding control units (*Required C*) and the control reports (*Required D*). 2. Library file – **HRLib.dll**
3. Your project folder with the library code and unittests.

~~The delivery date will be set~~ during the **February 2023 examination period**, as well as the date of the examination of the work. _____

WORKING GROUPS

Up to **3 people** (*according to the statement of the Alternative Labor group*)